

Tribhuvan University
Faculty of Management

Level: Bachelor

Program: Bachelor of Information Technology Management (BITM)

Course Title: Introduction to Computing and Information Technology

Course Code: ITM 101

Nature of the Course: Theory + Practical

Credit Value: 3

Semester: I

Total Hours: 48 (Th.) + 16(Pr.)

Course Description

This course provides a general overview of Information Technology (IT), which includes computer fundamentals, hardware, software, data management, networking, multimedia, and information systems. It also covers cybersecurity, IT ethics, and emerging technologies such as AI, blockchain, cloud computing, and IoT. Students will acquire practical skills and knowledge to apply IT in business, communication, education, and modern society with a focus on secure and ethical use of technology.

Course Objectives

By the end of this course, students will be able to

- Understand the fundamentals of computers, including types, structure, number systems, and data representation.
- Gain an acquaintance with hardware devices, mass storage media, I/O devices, and their management.
- Explore software types, operating systems, programming languages, and translators.
- Understand data administration concepts, databases, and the data processing life cycle.
- Gain knowledge of networking fundamentals, Internet technologies, protocols, and applications such as e-commerce and digital marketing.
- Gain awareness about cybersecurity, IT ethics, and professional standards in IT.
- Get familiar with emerging technologies like AI, blockchain, cloud computing, IoT, and digital transformation, and how they are applied in today's society and business.
- Apply IT knowledge to real-world problem-solving and decision-making at work and in daily life.

Course Details

Unit 1: Computer Fundamentals

5 LHs

Definition, Characteristics and Application of computers; Generations and Classification (based on working mechanism and size) of Computer; Architecture, Von Neumann model; Introduction

to Number Systems (binary, octal, decimal, hexadecimal) and conversion; Data representation (ASCII, Unicode)

Learning Outcomes: Define computers, their characteristics, and applications. (Remember); Classify computers based on generation, size, and functionality. (Understand); Explain the Von Neumann architecture and system components. (Understand); Convert numbers among binary, octal, decimal, and hexadecimal systems. (Apply); Describe data representation methods using ASCII and Unicode. (Understand)

Unit 2: Hardware & Peripherals

8 LHs

Introduction to Hardware; CPU, Components of CPU; Primary memory and its Types (Registers, Cache Memory, ROM and its types, RAM and its types); Secondary storage(Hard Disk Drives, Magnetic Tapes, Optical disc, SSDs, USB Flash Drives and Memory Cards); Memory hierarchy; System Buses and its type; Input devices(Keyboard, Mouse, Scanner, Microphone, Webcam, Joystick, Touch panel); Output devices (Monitor(CRT, LCD,LED, Plasma), Printer(Impact(Dot Matrix) and Non-Impact(Ink Jet and Laser)), Speakers); Motherboards (Definition and functions).

Learning Outcomes: Identify major hardware components and their functions. (Remember); Differentiate between types of memory and storage devices. (Analyze); Explain the role of input/output devices and system buses. (Understand); Illustrate the structure and function of a motherboard. (Apply); Compare performance characteristics of storage and display devices. (Evaluate)

Unit 3: Software

6 LHs

Introduction to Software, Types of Software (System, Application, and Utility Software); Introduction to Operating Systems; Function of and Types(On the basis of User interface, Function and Number of Users) of Operating System; Open-source vs proprietary software; Programming languages (machine level, Assembly level and High Level), Scripting Language (Server and client side), Language Translator (assembler, compiler, interpreter)

Learning Outcomes: Define software and distinguish between system, application, and utility software. (Remember, Understand); Explain the functions and types of operating systems. (Understand); Differentiate between open-source and proprietary software. (Analyze); Classify programming languages and translators. (Understand); Apply appropriate software tools for given computing tasks. (Apply)

Unit 4: Fundamental concept of Data Management

4 LHs

Introduction to Data and Information (Concept of data, information, and knowledge); Data representation and file organization; Database; Database approach vs. file-based system; DBMS, Components of DBMS, Data models (Introduction to Hierarchical, Network, Relational, and

Object-Oriented); Introduction to Data Processing, Steps of Data processing (Data collection, Data preparation, Data input, Data processing, Data output, Data storage).

Learning Outcomes: Define data, information, and knowledge. (Remember); Explain database concepts and components of DBMS. (Understand); Differentiate between file-based and database approaches. (Analyze); Describe the steps in the data processing cycle. (Understand); Apply data organization concepts in simple database examples. (Apply)

Unit 5: Networking & Internet

7 LHs

Network fundamentals (Definition of a Network, Purpose of Networking, Uses of Networking, Advantages and disadvantages of Networking), Components of Network; Transmission Media(Introduction to Guided and Unguided media); Network Topologies(Bus, Ring, Star, Mesh and Hybrid); Types of Networks (Based on size and architecture), Overview of OSI and TCP/IP model; Protocols(Introduction to IP, HTTP, SMTP, POP, IMAP, FTP, Telnet, SSL, HTTPS); IP address (Introduction of IPV4 and IPV6); Introduction to WWW and DNS; Internet, Intranet, Extranet; Web pages and Websites; Search Engines; Email; Introduction to E-commerce, E-governance, Digital Marketing and Social Media; Censorship and privacy issues.

Learning Outcomes: Explain basic networking concepts and their purposes. (Understand); Identify network components, media, and topologies. (Remember); Differentiate types of networks and network architectures. (Analyze); Describe common Internet protocols and addressing schemes. (Understand); Apply networking concepts in Internet and e-commerce contexts. (Apply); Discuss privacy, censorship, and security concerns online. (Evaluate)

Unit 6: Multimedia

3 LHs

Introduction to Multimedia (Definition, Characteristics and Importance), Multimedia Applications, Elements of a multimedia (Text, Image, Audio, Video and Animation); Image File Format (TIFF, JPEG, GIF, PNG)

Learning Outcomes: Define multimedia and its components. (Remember); Describe multimedia applications in communication and education. (Understand); Identify common image and video file formats. (Remember); Apply multimedia concepts in basic digital content creation. (Apply)

Unit 7: Introduction to IT & Information Systems

3 LHs

Definition and concept of IT; Components of IT; Importance of IT in modern society; Role of IT in communication, education, business, and government; Definition and concept of Information System (IS); Relationship between IT and IS; Types of Information Systems (TPS, MIS, DSS, ESS); Role of IS in business decision-making.

Learning Outcomes: Define Information Technology and its components. (Remember); Explain the role of IT in business, education, and governance. (Understand); Describe types of information systems and their business roles. (Understand); Analyze the relationship between IT and IS in organizational decision-making. (Analyze)

Unit 8: Cybersecurity

4 LHs

Computer Crime; Malware (Virus, Worm, Trojan, spyware, ransomware, adware); Type of attack (Interception, Interruption, Modification and Fabrication); Antivirus and Anti-malware; Hacking and Cracking; Software Piracy and Anti-Piracy; Security goal (CIA Triad); Basic of cryptography (Encryption and Decryption); Introduction to Firewall and its application; Introduction and need of Cyber law.

Learning Outcomes: Identify common cyber threats, attacks, and malware. (Remember); Explain the principles of the CIA triad and cryptography basics. (Understand); Apply basic security measures and use antivirus/firewall tools. (Apply); Discuss cyber laws and ethical security practices. (Evaluate)

Unit 9: IT Ethics

3 LHs

Intellectual Property (Copyright, Patent, Trade Secrets); Intellectual Property Issues (Plagiarism, Reverse Engineering); IT ethics, professional codes of conduct; Ethics in Information Technology; Encouraging Ethical Use of IT Resources among Users; Ethical Hacking

Learning Outcomes: Define intellectual property and ethical principles in IT. (Remember); Explain plagiarism, software piracy, and reverse engineering issues. (Understand); Analyze professional codes of conduct and ethical dilemmas in IT. (Analyze); Promote responsible and ethical use of IT resources. (Evaluate/Create)

Unit 10: Emerging Technologies & Digital Transformation

5 LHs

Overview of: AI, Machine Learning, Blockchain, VR/AR; Cloud Computing, Green Computing; Digital entrepreneurship, Emerging Trends in Data Management, Big Data and data analytics basics, Cloud-based data storage and management, Data governance, Data warehouses, data mining, cloud databases, Hybrid Storage:(Solid-State Hybrid Drives), IoT and Its integration in business, Smart City.

Learning Outcomes: Identify key emerging technologies such as AI, IoT, blockchain, and cloud computing. (Remember); Explain how emerging technologies transform business and society. (Understand); Discuss trends in data analytics, big data, and green computing. (Evaluate); Apply digital transformation concepts to real-world business cases. (Apply/Create)

Pedagogical Strategies

- Lectures with demonstration
- Hands on lab sessions
- Problem-based learning
- Guest lectures from tech industry experts
- Continuous assessment and feedback
- Multimedia presentations to visualize concepts
- Mini Project

Mode of Delivery

- Lecture sessions (Theory)
- Demonstration
- Laboratory work (Practical)
- Mini project

Internal Assessment Methods and Types

Assessment Type	Weightage	Details
Class participation & attendance	10%	Contribution to discussions, engagement in class activities
Quizzes/short tests	15%	Periodic quizzes to assess comprehension
Practical/Project	20%	Lab sessions and mini project
Mid-term examination	25%	Written test
Pre-board examination	30%	Comprehensive written test covering all units

External Assessment Methods and Types

Out of the total 60% allocated for final assessment, 40% will be assigned to the written/board examination to evaluate students' abilities in remembering, understanding, applying, analyzing, evaluating, and creating. The remaining 20% will be assigned to the final practical and competency.

Mapping Course: Learning Outcomes and Program Learning

Course Learning Objective (CLO) Dimensions	Knowledge (K)	Skills (S)	Competence (C)
Total Learning	35%	40%	25%

Laboratory / Practical Component**16 LHs**

Hands-on exercises with Microsoft Office

Microsoft Word

Navigate the Word interface and manage documents (create, save, format).

Type, copy, paste, and format text.

Adjust document layout (margins, orientation, headers/footers, page numbers).

Create and format tables, insert images, shapes, charts, hyperlinks, and symbols.

Apply styles, themes, and templates.

Use reviewing tools (spell check, track changes, comments).

Generate a Table of Contents.

Apply all these features in creating reports, letters, and formatted documents.

Microsoft Excel

Navigate the spreadsheet interface, workbooks, and worksheets.

Enter and manage data efficiently.

Format cells and work with numbers, text, and dates.

Apply basic formulas: SUM, AVERAGE, MIN, MAX.

Use advanced functions: IF, VLOOKUP, COUNTIF, CONCATENATE, AND, OR, IFAND, IFOR

Visualize data using charts and conditional formatting.

Sort, filter, and organize data using tables.

Apply practical exercises like preparing budgets, sales reports, Marksheet, etc.

Microsoft PowerPoint

Create and design slides using layouts, themes, and design principles.

Add and format text, images, charts, SmartArt, tables, and Hyperlink.

Enhance presentations with animations, transitions, and multimedia (audio/video).

Apply techniques for effective presentations and collaboration.

Export slides and presentations.

Complete practical exercises by creating full presentations.

References

Turban, E., Pollard, C., & Wood, G. (2018). *Information technology for management*. Wiley.

Norton, P. (2008). *Introduction to computers*. Tata McGraw-Hill.

Morley, D., & Parker Charles (2015). *Understanding Computers: Today and Tomorrow, Comprehensive 15th Edition*. Cengage Learning.

Laudon, K. C., & Laudon, J. P. (2012). *Management information systems. Person education*.

Laudon, K. C., & Traver, C. G. (2016). *E-commerce: Business, technology, society*. Pearson.